

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Hello, this is a mock exam for studying for the **quiz** that will be on 9/25/2026 for **MTH 181-03: Discrete Mathematics I**. As a reminder, you may share this document with others since you're helping spread learning, but please credit me for making this, since I'm making this in my very limited free time. Good luck studying.

**Note:** Have an understanding in **Sets 1.1 & 1.2**, but please try to focus more on **Set 2.1** for the upcoming quiz.

### Set 2.1

1. Provide an **example** of each of the laws we have learned. Please remember the following:

Let  $p$ ,  $q$ , and  $r$  be statement variables.

Let  $t$  represent a **tautology**.

Let  $c$  represent a **contradiction**.

- (a) Commutative laws: \_\_\_\_\_
- (b) Associative laws: \_\_\_\_\_
- (c) Distributive laws: \_\_\_\_\_
- (d) Identity laws: \_\_\_\_\_
- (e) Negation laws: \_\_\_\_\_
- (f) Double negative law: \_\_\_\_\_
- (g) Idempotent laws: \_\_\_\_\_
- (h) Universal bound laws: \_\_\_\_\_
- (i) De Morgan's laws: \_\_\_\_\_
- (j) Absorption laws: \_\_\_\_\_
- (k) Negations of  $t$  and  $c$ : \_\_\_\_\_

2. Argument to Common Form

Original Argument

If I pass this final exam, then I won't be a disappointment to my family.  
I'm a disappointment to my family.  
Therefore, it is not the case that I passed my final exam.

Common Form

3. Make a truth table for the following statement:  $\neg(p \vee q) \wedge (\neg p \wedge \neg q)$

| $p$ | $q$ | $\neg p$ | $\neg q$ | $p \vee q$ | $\neg(p \vee q)$ | $\neg p \wedge \neg q$ | $\neg(p \vee q) \wedge (\neg p \wedge \neg q)$ |
|-----|-----|----------|----------|------------|------------------|------------------------|------------------------------------------------|
| T   | T   |          |          |            |                  |                        |                                                |
| T   | F   |          |          |            |                  |                        |                                                |
| F   | T   |          |          |            |                  |                        |                                                |
| F   | F   |          |          |            |                  |                        |                                                |

4. Truth Table: Are  $(p \wedge q)$  and  $(\neg p \vee \neg q)$  logically equivalent?

| $p$ | $q$ | $\neg p$ | $\neg q$ | $p \wedge q$ | $\neg p \vee \neg q$ |
|-----|-----|----------|----------|--------------|----------------------|
| T   | T   |          |          |              |                      |
| T   | F   |          |          |              |                      |
| F   | T   |          |          |              |                      |
| F   | F   |          |          |              |                      |

☐ True

☐ False

5. Truth Table: Are  $(p \vee (q \wedge r))$  and  $(p \vee q) \wedge (p \vee r)$  logically equivalent?

| $p$ | $q$ | $r$ | $q \wedge r$ | $p \vee q$ | $p \vee r$ | $p \vee (q \wedge r)$ | $(p \vee q) \wedge (p \vee r)$ |
|-----|-----|-----|--------------|------------|------------|-----------------------|--------------------------------|
| T   | T   | T   |              |            |            |                       |                                |
| T   | T   | F   |              |            |            |                       |                                |
| T   | F   | T   |              |            |            |                       |                                |
| T   | F   | F   |              |            |            |                       |                                |
| F   | T   | T   |              |            |            |                       |                                |
| F   | T   | F   |              |            |            |                       |                                |
| F   | F   | T   |              |            |            |                       |                                |
| F   | F   | F   |              |            |            |                       |                                |

☐ True

☐ False

6. Truth Table: Is  $p \vee \neg p$  a tautology, a contradiction, or neither?

| $p$ | $\neg p$ | $p \vee \neg p$ |
|-----|----------|-----------------|
|     |          |                 |
|     |          |                 |

☐ Tautology

☐ Contradiction

☐ Neither

7. Please provide reasoning to prove this logical equivalence.

$$\neg(p \vee q) \wedge (\neg(\neg p \vee q)) \equiv c$$

|                                                                     |                     |
|---------------------------------------------------------------------|---------------------|
| $\equiv (\neg p \wedge \neg q) \wedge (\neg(\neg p) \wedge \neg q)$ | by De Morgan's laws |
| $\equiv (\neg p \wedge \neg q) \wedge (p \wedge \neg q)$            |                     |
| $\equiv (\neg q \wedge \neg p) \wedge (\neg q \wedge p)$            |                     |
| $\equiv \neg q \wedge (\neg p \wedge p)$                            |                     |
| $\equiv \neg q \wedge (p \wedge \neg p)$                            |                     |
| $\equiv \neg q \wedge c$                                            |                     |
| $\equiv c$                                                          |                     |

8. Prove De Morgan's Laws by making a truth table.

(a)  $\neg(p \wedge q) \equiv \neg p \vee q$

(b)  $\neg(p \vee q) \equiv \neg p \wedge \neg q$

9. Rewrite the following inequalities symbolically:

Let  $p = "x > -19"$ ,  $q = "x < 9"$ ,  $r = "x = -19"$ , and  $s = "x = 9"$

(a)  $x \geq -19$

(b)  $-19 < x < 9$

(c)  $-19 \leq x < 9$

(d)  $-19 \leq x \leq 9$